12 — Appendix: Research

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1. Position

This appendix indexes the **external, citable findings** that shaped the HelixVPN master specification. Every research angle was investigated with live web access (2026-06-25) and is recorded in its own dossier under `v09-research/`. This README is a navigable summary; the dossiers and `11-deep-research-appendix.md` retain the full sources and verbatim URLs.

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2. Research angles

Dossier	Topic	Key finding	Surfaced in
`research-masque.md`	MASQUE / CONNECT-UDP / HTTP Datagrams	WG-over-HTTP/3 is a stack of stable RFCs (9298, 9297, 9221, 9220); Mullvad shipped QUIC obfuscation in 2025.9 using `quinn` + MASQUE.	D1 Camp A, `01-data-plane`, `08-phase2-parity-wbs.md`
`research-hysteria2.md`	Hysteria2 + Salamander / Gecko, sing-box	Hysteria2 v2.9.2 adds Gecko handshake fragmentation; `masque-go` is turnkey in Go; sing-box is a universal multiplexer but **not adopted**.	D1 Camp B, Phase-2 transport set, source gap G2
`research-wireguard.md`		Noise IK 1-RTT; fixed primitives (Curve25519,	`01-data-plane`, `v02-data-plane/`

Dossier	Topic	Key finding	Surfaced in
	WireGuard protocol + implementations	ChaCha20Poly1305, BLAKE2s, HKDF, SipHash24, XChaCha20Poly1305); exact wire sizes confirmed (148/92/64/32 B).	`wireguard-core.md`
`research-mullvad.md`	Mullvad parity feature set	DAITA v1/v2 built on maybenot; constant packet size + cover traffic + pattern distortion; server-driven defenses in v2.	`04-security-privacy-pki`, Phase-2 DAITA track
`research-daita_test.md`	maybenot + VPN testing rigs	maybenot 2.2.2 core, MIT, used by Mullvad; Linux netns + nftables + tc netem is the standard test harness.	DAITA design, `10-testing-acceptance-and-qa.md`
`research-pki_pq_nat.md`	PKI, post-quantum, NAT traversal	SPIFFE/SPIRE-style short-lived mTLS over WG; ML-KEM hybrid PSK (Mullvad/ Rosenpass); Tailscale-style STUN+hole-punch+DERP relay.	`04-security-privacy-pki`, `08-phase2-parity-wbs.md`
`research-ios_android.md`	Mobile native constraints	iOS NEPacketTunnelProvider limit ≈ 50 MiB (unofficial), plan to ~12–15 MB working set; Rust staticlib size optimizations mandatory.	`03-client-core-and-ui`, G3 gate
`research-go_cp.md`	Go control-plane stack	Gin 1.12.0, Connect-RPC 1.20.0, sqlc 1.31.1, protovalidate v1.0 — versions and best practices.	`02-control-plane`, `08-api-contracts`
`research-flutter_ffi.md`	Rust ↔ Flutter / native bridges	flutter_rust_bridge 2.12.0 recommended; UniFFI-Dart experimental; per-platform staticlib/cdylib packaging.	`03-client-core-and-ui`
`research-podman_k8s.md`	Rootless Podman, quadlets, K8s	Quadlet is core in Podman 5.x; rootless unit search paths; `DropCapability=ALL` +	`07-infrastructure-devops`

Dossier	Topic	Key finding	Surfaced in
		`AddCapability=NET_ADMIN NET_RAW` + `/dev/net/tun`.	

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3. Cross-doc synthesis

The high-level synthesis in `v09-research/_SYNTHESIS.md` distills the 16 original source documents into:

- The settled product floor (self-hostable overlay network + privacy-VPN front end).
- The near-unanimous stack floor (Go + Gin + Postgres + Redis + rootless Podman; WireGuard; Flutter).
- The eight key decisions D1–D8 (obfuscation transport, client-core language, event bus, subnet collision, edge language, transport topology, MVP ambition, licensing).
- The four-phase roadmap used by the WBS docs.

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4. Decision coverage check

All eight decisions are surfaced explicitly somewhere in `final/`:

Decision	Camp A	Camp B	Surfaced in
D1 obfuscation	MASQUE/ QUIC (Mullvad parity)	Hysteria2 + Salamander	`01-data-plane`, `08-phase2-parity-wbs.md`, `11-deep-research-appendix.md`
D2 client core	Rust + Flutter	Go + Flutter	`03-client-core-and-ui.md`, `11-deep-research-appendix.md`
D3 event bus	Redis Streams → NATS	NATS from start	`02-control-plane.md`, `08-phase2-parity-wbs.md`
D4 subnet collision	IPv6 ULA /48 + 4via6	CGNAT 100.64/10	`02-control-plane.md`, `08-phase2-parity-wbs.md`
D5 edge language	Rust (quinn+h3)	Go (quic-go)	`01-data-plane.md`, G4 gate
D6 transport topology	single protocol	asymmetric per-leg	`01-data-plane.md`, `11-deep-research-appendix.md`
D7 ambition			

Decision	Camp A	Camp B	Surfaced in
	lean tunnel-first	full Connectivity-OS	`00-product-scope-and-principles.md`, `SPECIFICATION.md`
D8 licensing	source-available + commercial	pure OSS	`SPECIFICATION.md` §9 (pre-Phase-3)

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5. Honest boundaries

- This appendix is an **index**; every factual claim is backed by the cited dossier and its sources.
- Some values (e.g. exact iOS NE memory limit, future OpenDesign exporter shapes) are marked `UNVERIFIED` in the source dossiers per §11.4.6.
- Tool versions are pinned to the access date (2026-06-25) and should be re-verified before procurement.

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6. Cross-references

- Deep-research appendix → `../../11-deep-research-appendix.md`
- Cross-doc synthesis → `../../v09-research/_SYNTHESIS.md`
- Source coverage ledger → 99 — Source Coverage Ledger

Sources: `11-deep-research-appendix.md`, `v09-research/*.md`.